

Finding Groups in Data

An Introduction to Cluster Analysis

LEONARD KAUFMAN

Vrije Universiteit Brussel,
Brussels, Belgium

PETER J. ROUSSEEUW

Universitaire Instelling Antwerpen,
Antwerp, Belgium



A JOHN WILEY & SONS, INC., PUBLICATION

This Page Intentionally Left Blank

Finding Groups in Data

This Page Intentionally Left Blank

Finding Groups in Data

An Introduction to Cluster Analysis

LEONARD KAUFMAN

Vrije Universiteit Brussel,
Brussels, Belgium

PETER J. ROUSSEEuw

Universitaire Instelling Antwerpen,
Antwerp, Belgium



A JOHN WILEY & SONS, INC., PUBLICATION

Copyright © 1990, 2005 by John Wiley & Sons, Inc. All rights reserved.

Published by John Wiley & Sons, Inc., Hoboken, New Jersey.

Published simultaneously in Canada.

No part of this publication may be reproduced, stored in a retrieval system or transmitted in any form or by any means, electronic, mechanical, photocopying, recording, scanning or otherwise, except as permitted under Sections 107 or 108 of the 1976 United States Copyright Act, without either the prior written permission of the Publisher, or authorization through payment of the appropriate per-copy fee to the Copyright Clearance Center, 222 Rosewood Drive, Danvers, MA 01923, (978) 750-8400, fax (978) 750-4470. Requests to the Publisher for permission should be addressed to the Permissions Department, John Wiley & Sons, Inc., 111 River Street, Hoboken, NJ 07030, (201) 748-6011, fax (201) 748-6008.

Limit of Liability/Disclaimer of Warranty: While the publisher and author have used their best efforts in preparing this book, they make no representation or warranties with respect to the accuracy or completeness of the contents of this book and specifically disclaim any implied warranties of merchantability or fitness for a particular purpose. No warranty may be created or extended by sales representatives or written sales materials. The advice and strategies contained herein may not be suitable for your situation. You should consult with a professional where appropriate. Neither the publisher nor author shall be liable for any loss of profit or any other commercial damages, including but not limited to special, incidental, consequential, or other damages.

For general information on our other products and services please contact our Customer Care Department within the U.S. at 877-762-2974, outside the U.S. at 317-572-3993 or fax 317-572-4002.

Wiley also publishes its books in a variety of electronic formats. Some content that appears in print, however, may not be available in electronic format.

Library of Congress Cataloging-in-Publication is available.

ISBN 0-471-73578-7

Printed in the United States of America.

10 9 8 7 6 5 4 3 2 1

To Our Parents And To Jean Haezendonck

This Page Intentionally Left Blank

Preface

Cluster analysis is a very practical subject. Some 30 years ago, biologists and social scientists began to look for systematic ways to find groups in their data. Because computers were becoming available, the resulting algorithms could actually be implemented. Nowadays clustering methods are applied in many domains, including artificial intelligence and pattern recognition, chemometrics, ecology, economics, the geosciences, marketing, medical research, political science, psychometrics, and many more. This has led to a lot of different methods, and articles on clustering have appeared not only in statistical journals but also in periodicals of all these domains. Clustering is known under a variety of names, such as numerical taxonomy and automatic data classification.

Our purpose was to write an applied book for the general user. We wanted to make cluster analysis available to people who do not necessarily have a strong mathematical or statistical background. Rather than giving an extensive survey of clustering methods, leaving the user with a bewildering multitude of methods to choose from, we preferred to select a few methods that together can deal with most applications. This selection was based on a combination of methodological aims (mainly robustness, consistency, and general applicability) and our own experience in applying clustering to a variety of disciplines.

The book grew out of several courses on cluster analysis that we taught in Brussels, Delft, and Fribourg. It was extensively tested as a textbook with students of mathematics, biology, economics, and political science. It is one of the few books on cluster analysis containing exercises. The first chapter introduces the main approaches to clustering and provides guidance to the choice between the available methods. It also discusses various types of data (including interval-scaled and binary variables, as well as similarity data) and explains how these can be transformed prior to the actual

clustering. The other six chapters each deal with a specific clustering method. These chapters all have the same structure. The first sections give a short description of the clustering method, explain how to use it, and discuss a set of examples. These are followed by two sections (marked with * because they may be skipped without loss of understanding) on the algorithm and its implementation, and on some related methods in the literature. The chapters are relatively independent (except for Chapter 3 which builds on Chapter 2), allowing instructors to cover only one chapter in a statistics course. Another advantage is that researchers can pick out the method they need for their current application, without having to read other chapters. (To achieve this structure, some things had to be repeated in the text.) Occasionally, we handed a single chapter to someone working on a particular problem.

Chapters 2, 3, and 4 are about partitioning methods, whereas Chapters 5, 6, and 7 cover hierarchical techniques. Whenever possible, we constructed methods that cannot only analyze data consisting of measurements (i.e., objects with variables) but also data consisting of dissimilarities between objects. (This excluded parametric approaches, such as those based on multivariate mixture distributions.) We also wanted the methods to be consistent for large data sets. All the selected methods are of the L_1 type, which means that they minimize sums of dissimilarities (rather than sums of *squared* dissimilarities, as in the classical nonrobust methods). Some of the methods are new, such as the approach for partitioning large data sets and the L_1 method for fuzzy clustering. Also, the clusterings are accompanied by graphical displays (called silhouettes and banners) and corresponding quality coefficients, which help the user to select the number of clusters and to see whether the method has found groups that were actually present in the data.

Current statistical software packages contain only a few clustering techniques and they are not the more modern methods. This forced us to write new programs, the use of which is described in the book. Their present version is for IBM-PC and compatible machines, but the source codes are very portable and have run on several types of mainframes without problems. The programs (together with their sources and the data sets used in the book) are available on floppy disks by writing to the authors. The programs are also being integrated in the workstation package S-PLUS of Statistical Sciences, Inc., P.O. Box 85625, Seattle, WA 98145-1625.

We are grateful to Frank Critchley, Jan de Leeuw, Gaetan Libert, Glenn Milligan, Frank Plastria, Marc Roubens, John Tukey, Bert van Zomeren, Howard Wainer, Michael Windham, and David Wishart for helpful suggestions and stimulating discussions on topics covered in this book and to

Etienne Trauwaert for contributing to Section 4 of Chapter 4. Valuable comments on the manuscript were given by Phil Hopke, Doug Martin, Marie-Paule Derde, and two reviewers. Annie De Schrijver was responsible for drawing several figures. Finally, we would like to thank our wives, Jacqueline and Lieve, for their patience and support.

LEONARD KAUFMAN
PETER J. ROUSSEEUW

Edegem, Belgium
August, 1989

This Page Intentionally Left Blank

Contents

1. Introduction	1
1. Motivation, 1	
2. Types of Data and How to Handle Them, 3	
2.1 Interval-Scaled Variables, 4	
2.2 Dissimilarities, 16	
2.3 Similarities, 20	
2.4 Binary Variables, 22	
2.5 Nominal, Ordinal, and Ratio Variables, 28	
2.6 Mixed Variables, 32	
3. Which Clustering Algorithm to Choose, 37	
3.1 Partitioning Methods, 38	
3.2 Hierarchical Methods, 44	
4. A Schematic Overview of Our Programs, 50	
5. Computing Dissimilarities with the Program DAISY, 52	
Exercises and Problems, 63	
2. Partitioning Around Medoids (Program PAM)	68
1. Short Description of the Method, 68	
2. How to Use the Program PAM, 72	
2.1 Interactive Use and Input, 72	
2.2 Output, 80	
2.3 Missing Values, 88	
3. Examples, 92	
*4. More on the Algorithm and the Program, 102	
4.1 Description of the Algorithm, 102	
4.2 Structure of the Program, 104	

*5. Related Methods and References, 108	
5.1 The k -Medoid Method and Optimal Plant Location, 108	
5.2 Other Methods Based on the Selection of Representative Objects, 110	
5.3 Methods Based on the Construction of Central Points, 111	
5.4 Some Other Nonhierarchical Methods, 116	
5.5 Why Did We Choose the k -Medoid Method?, 117	
5.6 Graphical Displays, 119	
Exercises and Problems, 123	
3. Clustering Large Applications (Program CLARA)	126
1. Short Description of the Method, 126	
2. How to Use the Program CLARA, 127	
2.1 Interactive Use and Input, 127	
2.2 Output, 130	
2.3 Missing Values, 134	
3. An Example, 139	
*4. More on the Algorithm and the Program, 144	
4.1 Description of the Algorithm, 144	
4.2 Structure of the Program, 146	
4.3 Limitations and Special Messages, 151	
4.4 Modifications and Extensions of CLARA, 153	
*5. Related Methods and References, 155	
5.1 Partitioning Methods for Large Data Sets, 155	
5.2 Hierarchical Methods for Large Data Sets, 157	
5.3 Implementing CLARA on a Parallel Computer, 160	
Exercises and Problems, 162	
4. Fuzzy Analysis (Program FANNY)	164
1. The Purpose of Fuzzy Clustering, 164	
2. How to Use the Program FANNY, 166	
2.1 Interactive Use and Input, 167	
2.2 Output, 170	
3. Examples, 175	
*4. More on the Algorithm and the Program, 182	
4.1 Description of the Algorithm, 182	
4.2 Structure of the Program, 188	

*5. Related Methods and References, 189	
5.1 Fuzzy k -Means and the MND2 Method, 189	
5.2 Why Did We Choose FANNY?, 191	
5.3 Measuring the Amount of Fuzziness, 191	
5.4 A Graphical Display of Fuzzy Memberships, 195	
Exercises and Problems, 197	
5. Agglomerative Nesting (Program AGNES)	199
1. Short Description of the Method, 199	
2. How to Use the Program AGNES, 208	
2.1 Interactive Use and Input, 208	
2.2 Output, 209	
3. Examples, 214	
*4. More on the Algorithm and the Program, 221	
4.1 Description of the Algorithm, 221	
4.2 Structure of the Program, 223	
*5. Related Methods and References, 224	
5.1 Other Agglomerative Clustering Methods, 224	
5.2 Comparing Their Properties, 238	
5.3 Graphical Displays, 243	
Exercises and Problems, 250	
6. Divisive Analysis (Program DIANA)	253
1. Short Description of the Method, 253	
2. How to Use the Program DIANA, 259	
3. Examples, 263	
*4. More on the Algorithm and the Program, 271	
4.1 Description of the Algorithm, 271	
4.2 Structure of the Program, 272	
*5. Related Methods and References, 273	
5.1 Variants of the Selected Method, 273	
5.2 Other Divisive Techniques, 275	
Exercises and Problems, 277	
7. Monothetic Analysis (Program MONA)	280
1. Short Description of the Method, 280	
2. How to Use the Program MONA, 283	

- 2.1 Interactive Use and Input, 284
- 2.2 Output, 287
- 3. Examples, 290
- *4. More on the Algorithm and the Program, 298
 - 4.1 Description of the Algorithm, 298
 - 4.2 Structure of the Program, 301
- *5. Related Methods and References, 304
 - 5.1 Association Analysis, 304
 - 5.2 Other Monothetic Divisive Algorithms for Binary Data, 307
 - 5.3 Some Other Divisive Clustering Methods, 308
- Exercises and Problems, 310

APPENDIX 312

- 1. Implementation and Structure of the Programs, 312
- 2. Running the Programs, 313
- 3. Adapting the Programs to Your Needs, 316
- 4. The Program CLUSPLOT, 318

References 320

Author Index 332

Subject Index 335

Finding Groups in Data

This Page Intentionally Left Blank

CHAPTER 1

Introduction

1 MOTIVATION

Cluster analysis is the art of finding groups in data. To see what is meant by this, let us look at Figure 1. It is a plot of eight objects, on which two variables were measured. For instance, the weight of an object might be displayed on the horizontal axis and its height on the vertical one. Because this example contains only two variables, we can investigate it by merely looking at the plot.

In this small data set there are clearly two distinct groups of objects, namely {TIN, TAL, KIM, ILA} and {LIE, JAC, PET, LEO}. Such groups are called *clusters*, and to discover them is the aim of cluster analysis. Basically, one wants to form groups in such a way that objects in the same group are similar to each other, whereas objects in different groups are as dissimilar as possible.

The classification of similar objects into groups is an important human activity. In everyday life, this is part of the learning process: A child learns to distinguish between cats and dogs, between tables and chairs, between men and women, by means of continuously improving subconscious classification schemes. (This explains why cluster analysis is often considered as a branch of pattern recognition and artificial intelligence.) Classification has always played an essential role in science. In the eighteenth century, Linnaeus and Sauvages provided extensive classifications of animals, plants, minerals, and diseases (for a recent survey, see Holman, 1985). In astronomy, Hertzsprung and Russell classified stars in various categories on the basis of two variables: their light intensity and their surface temperature. In the social sciences, one frequently classifies people with regard to their behavior and preferences. In marketing, it is often attempted to identify market segments, that is, groups of customers with similar needs. Many more examples could be given in geography (clustering of regions), medicine

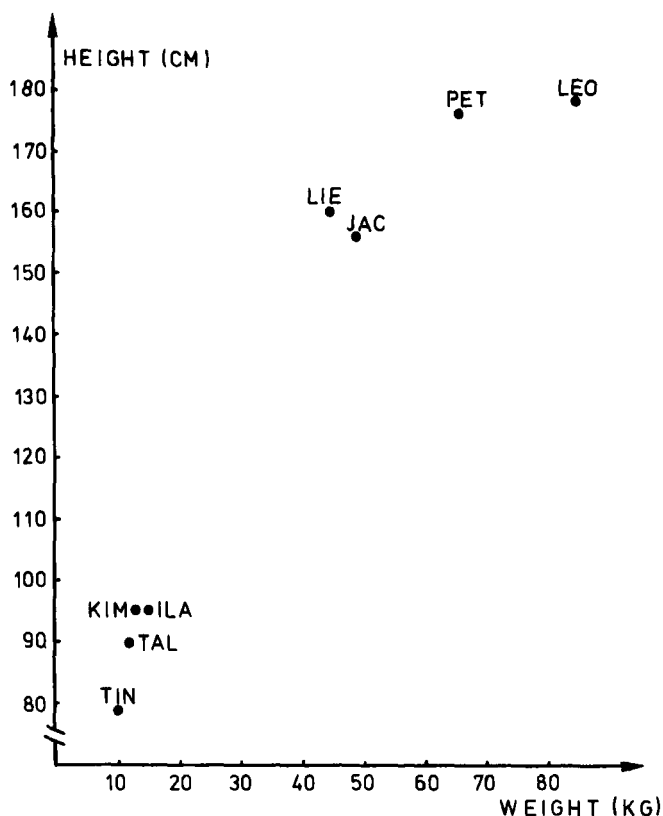


Figure 1 A plot of eight objects.

(incidence of specific types of cancer), chemistry (classification of compounds), history (grouping of archeological findings), and so on. Moreover, cluster analysis can be used not only to identify a structure already present in the data, but also to impose a structure on a more or less homogeneous data set that has to be split up in a "fair" way, for instance when dividing a country into telephone areas. Note that cluster analysis is quite different from discriminant analysis in that it actually establishes the groups, whereas discriminant analysis assigns objects to groups that were defined in advance.

In the past, clusterings were usually performed in a subjective way, by relying on the perception and judgment of the researcher. In the example of Figure 1, we used the human eye-brain system which is very well suited (through millenia of evolution) for classification in up to three dimensions.

However, the need to classify cases in more than three dimensions and the upcoming objectivity standards of modern science have given rise to so-called automatic classification procedures. Over the last 30 years, a wealth of algorithms and computer programs has been developed for cluster analysis. The reasons for this variety of methods are probably twofold. To begin with, automatic classification is a very young scientific discipline in vigorous development, as can be seen from the thousands of articles scattered over many periodicals (mostly journals of statistics, biology, psychometrics, computer science, and marketing). Nowadays, automatic classification is establishing itself as an independent scientific discipline, as witnessed by a full-fledged periodical (the *Journal of Classification*, first published in 1984) and the International Federation of Classification Societies (founded in 1985). The second main reason for the diversity of algorithms is that there exists no general definition of a cluster, and in fact there are several kinds of them: spherical clusters, drawn-out clusters, linear clusters, and so on. Moreover, different applications make use of different data types, such as continuous variables, discrete variables, similarities, and dissimilarities. Therefore, one needs different clustering methods in order to adapt to the kind of application and the type of clusters sought. Cluster analysis has become known under a variety of names, such as numerical taxonomy, automatic classification, botryology (Good, 1977), and typological analysis (Chandon and Pinson, 1981).

In this book, several algorithms are provided for transforming the data, for performing cluster analysis, and for displaying the results graphically. Section 2 of this introduction discusses the various types of data and what to do with them, and Section 3 gives a brief survey of the clustering methods contained in the book, with some guidelines as to which algorithm to choose. In particular the crucial distinction between partitioning and hierarchical methods is considered. In Section 4 a schematic overview is presented. In Section 5, it is explained how to use the program DAISY to transform your data.

2 TYPES OF DATA AND HOW TO HANDLE THEM

Our first objective is to study some types of data which typically occur and to investigate ways of processing the data to make them suitable for cluster analysis.

Suppose there are n objects to be clustered, which may be persons, flowers, words, countries, or whatever. Clustering algorithms typically operate on either of two input structures. The first represents the objects by means of p measurements or attributes, such as height, weight, sex, color,

and so on. These measurements can be arranged in an n -by- p matrix, where the rows correspond to the objects and the columns to the attributes. In Tucker's (1964) terminology such an objects-by-variables matrix is said to be *two-mode*, since the row and column entities are different. The second structure is a collection of proximities that must be available for all pairs of objects. These proximities make up an n -by- n table, which is called a *one-mode* matrix because the row and column entities are the same set of objects. We shall consider two types of proximities, namely dissimilarities (which measure how far away two objects are from each other) and similarities (which measure how much they resemble each other). Let us now have a closer look at the types of data used in this book by considering them one by one.

2.1 Interval-Scaled Variables

In this situation the n objects are characterized by p *continuous* measurements. These values are positive or negative real numbers, such as height, weight, temperature, age, cost, ..., which follow a linear scale. For instance, the time interval between 1905 and 1915 was equal in length to that between 1967 and 1977. Also, it takes the same amount of energy to heat an object of -16.4°C to -12.4°C as to bring it from 35.2°C to 39.2°C . In general it is required that intervals keep the same importance throughout the scale.

These measurements can be organized in an n -by- p matrix, where the rows correspond to the objects (or cases) and the columns correspond to the variables. When the f th measurement of the i th object is denoted by x_{if} (where $i = 1, \dots, n$ and $f = 1, \dots, p$) this matrix looks like

$$\begin{array}{c}
 \begin{array}{c} n \text{ objects} \end{array}
 \begin{bmatrix}
 x_{11} & \cdots & x_{1f} & \cdots & x_{1p} \\
 \vdots & & \vdots & & \vdots \\
 x_{i1} & \cdots & x_{if} & \cdots & x_{ip} \\
 \vdots & & \vdots & & \vdots \\
 x_{n1} & \cdots & x_{nf} & \cdots & x_{np}
 \end{bmatrix}
 \end{array}
 \quad (1)$$

For instance, consider the following real data set. For eight people, the weight (in kilograms) and the height (in centimeters) is recorded in Table 1. In this situation, $n = 8$ and $p = 2$. One could have recorded many more variables, like age and blood pressure, but as there are only two variables in this example it is easy to make a scatterplot, which corresponds to Figure 1

Table 1 Weight and Height of Eight People, Expressed in Kilograms and Centimeters

Name	Weight (kg)	Height (cm)
Ilan	15	95
Jacqueline	49	156
Kim	13	95
Lieve	45	160
Leon	85	178
Peter	66	176
Talia	12	90
Tina	10	78

given earlier. Note that the units on the vertical axis are drawn to the same size as those on the horizontal axis, even though they represent different physical concepts. The plot contains two obvious clusters, which can in this case be interpreted easily: the one consists of small children and the other of adults.

Note that other variables might have led to completely different clusterings. For instance, measuring the concentration of certain natural hormones might have yielded a clear-cut partition into three male and five female persons. By choosing still other variables, one might have found blood types, skin types, or many other classifications.

Let us now consider the effect of changing measurement units. If weight and height of the subjects had been expressed in pounds and inches, the results would have looked quite different. A pound equals 0.4536 kg and an inch is 2.54 cm. Therefore, Table 2 contains larger numbers in the column of weights and smaller numbers in the column of heights. Figure 2, although plotting essentially the same data as Figure 1, looks much flatter. In this figure, the relative importance of the variable "weight" is much larger than in Figure 1. (Note that Kim is closer to Ilan than to Talia in Figure 1, but that she is closer to Talia than to Ilan in Figure 2.) As a consequence, the two clusters are not as nicely separated as in Figure 1 because in this particular example the height of a person gives a better indication of adulthood than his or her weight. If height had been expressed in feet (1 ft = 30.48 cm), the plot would become flatter still and the variable "weight" would be rather dominant.

In some applications, changing the measurement units may even lead one to see a very different clustering structure. For example, the age (in years) and height (in centimeters) of four imaginary people are given in

**Table 2 Weight and Height of the Same Eight People,
But Now Expressed in Pounds and Inches**

Name	Weight (lb)	Height (in.)
Ilan	33.1	37.4
Jacqueline	108.0	61.4
Kim	28.7	37.4
Lieve	99.2	63.0
Leon	187.4	70.0
Peter	145.5	69.3
Talia	26.5	35.4
Tina	22.0	30.7

Table 3 and plotted in Figure 3. It appears that $\{A, B\}$ and $\{C, D\}$ are two well-separated clusters. On the other hand, when height is expressed in feet one obtains Table 4 and Figure 4, where the obvious clusters are now $\{A, C\}$ and $\{B, D\}$. This partition is completely different from the first because each subject has received another companion. (Figure 4 would have been flattened even more if age had been measured in days.)

To avoid this dependence on the choice of measurement units, one has the option of standardizing the data. This converts the original measurements to unitless variables. First one calculates the mean value of variable

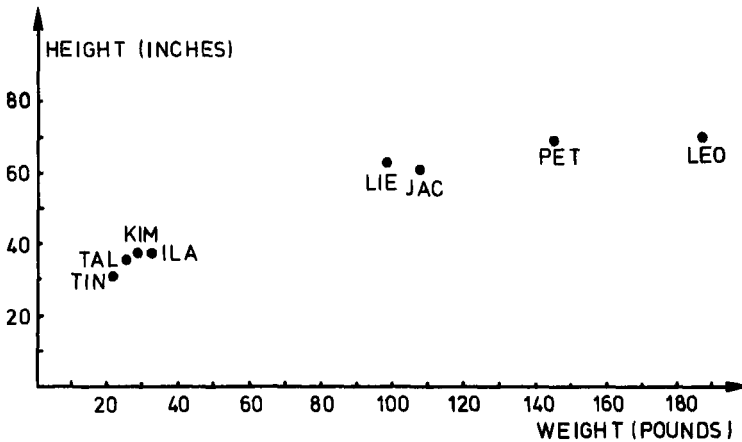
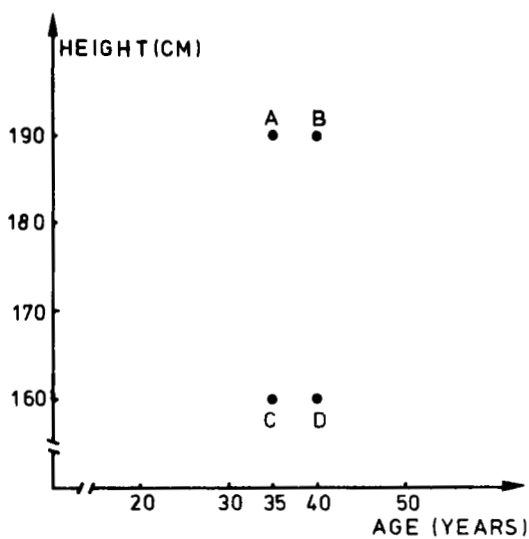


Figure 2 Plot corresponding to Table 2.

Table 3 Age (in years) and Height (in centimeters) of Four People

Person	Age (yr)	Height (cm)
A	35	190
B	40	190
C	35	160
D	40	160

**Figure 3** Plot of height (in centimeters) versus age for four people.**Table 4** Age (in years) and Height (in feet) of the Same Four People

Person	Age (yr)	Height (ft)
A	35	6.2
B	40	6.2
C	35	5.2
D	40	5.2

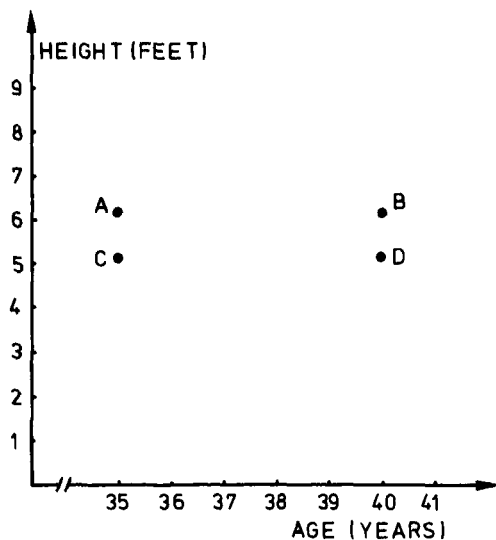


Figure 4 Plot of height (in feet) versus age for the same four people.

f , given by

$$m_f = \frac{1}{n}(x_{1f} + x_{2f} + \cdots + x_{nf}) \quad (2)$$

for each $f = 1, \dots, p$. Then one computes a measure of the dispersion or “spread” of this f th variable. Traditionally, people use the standard deviation

$$\text{std}_f = \sqrt{\frac{1}{n-1} \{ (x_{1f} - m_f)^2 + (x_{2f} - m_f)^2 + \cdots + (x_{nf} - m_f)^2 \}}$$

for this purpose. However, this measure is affected very much by the presence of outlying values. For instance, suppose that one of the x_{if} has been wrongly recorded, so that it is much too large. In this case std_f will be unduly inflated, because $x_{if} - m_f$ is squared. Hartigan (1975, p. 299) notes that one needs a dispersion measure that is not too sensitive to outliers. Therefore, from now on we will use the *mean absolute deviation*

$$s_f = \frac{1}{n} \{ |x_{1f} - m_f| + |x_{2f} - m_f| + \cdots + |x_{nf} - m_f| \} \quad (3)$$

where the contribution of each measurement x_{if} is proportional to the absolute value $|x_{if} - m_f|$. This measure is more robust (see, e.g., Hampel

et al., 1986) in the sense that one outlying observation will not have such a large influence on s_f . (Note that there exist estimates that are much more robust, but we would like to avoid a digression into the field of robust statistics.)

Let us assume that s_f is nonzero (otherwise variable f is constant over all objects and must be removed). Then the standardized measurements are defined by

$$z_{if} = \frac{x_{if} - m_f}{s_f} \quad (4)$$

and sometimes called *z-scores*. They are unitless because both the numerator and the denominator are expressed in the same units. By construction, the z_{if} have mean value zero and their mean absolute deviation is equal to 1. When applying standardization, one forgets about the original data (1) and uses the new data matrix

$$\begin{array}{cc} & \text{variables} \\ \text{objects} & \begin{bmatrix} z_{11} & \cdots & z_{1f} & \cdots & z_{1p} \\ \vdots & & \vdots & & \vdots \\ z_{i1} & \cdots & z_{if} & \cdots & z_{ip} \\ \vdots & & \vdots & & \vdots \\ z_{n1} & \cdots & z_{nf} & \cdots & z_{np} \end{bmatrix} \end{array} \quad (5)$$

in all subsequent computations. The advantage of using s_f rather than std_f in the denominator of (4) is that s_f will not be blown up so much in the case of an outlying x_{if} , and hence the corresponding z_{if} will still “stick out” so the i th object can be recognized as an outlier by the clustering algorithm, which will typically put it in a separate cluster. [This motivation differs from that of Milligan and Cooper (1988), who recommended the use of a nonrobust denominator.]

The preceding description might convey the impression that standardization would be beneficial in all situations. However, it is merely an option that may or may not be useful in a given application. Sometimes the variables have an absolute meaning, and should not be standardized (for instance, it may happen that several variables are expressed in the same units, so they should not be divided by different s_f). Often standardization dampens a clustering structure by reducing the large effects because the variables with a big contribution are divided by a large s_f .

For example, let us standardize the data of Table 3. The mean age equals $m_1 = 37.5$ and the mean absolute deviation of the first variable works out to be $s_1 = \{2.5 + 2.5 + 2.5 + 2.5\}/4 = 2.5$. Therefore, standardization converts age 40 to +1 and age 35 to -1. Analogously, $m_2 = 175$ cm and

Table 5 Standardized Age and Height of the Same Four People

Person	Variable 1	Variable 2
A	-1.0	1.0
B	1.0	1.0
C	-1.0	-1.0
D	1.0	-1.0

$s_2 = \{15 + 15 + 15 + 15\}/4 = 15$ cm, so 190 cm is standardized to +1 and 160 cm to -1. The resulting data matrix, which is unitless, is given in Table 5. Note that the new averages are zero and that the mean deviations equal 1. Of course, standardizing Table 4 would yield exactly the same result, so Table 5 is the standardized version of both Tables 3 and 4. Even when the data are converted to very strange units (such as the proverbial fortnights and furlongs), standardization will always yield the same numbers. However, plotting the values of Table 5 in Figure 5 does not give a very exciting result. Figure 5 looks like an intermediate between Figures 3 and 4 and shows no clustering structure because the four points lie at the vertices of a square. One could say that there are four clusters, each

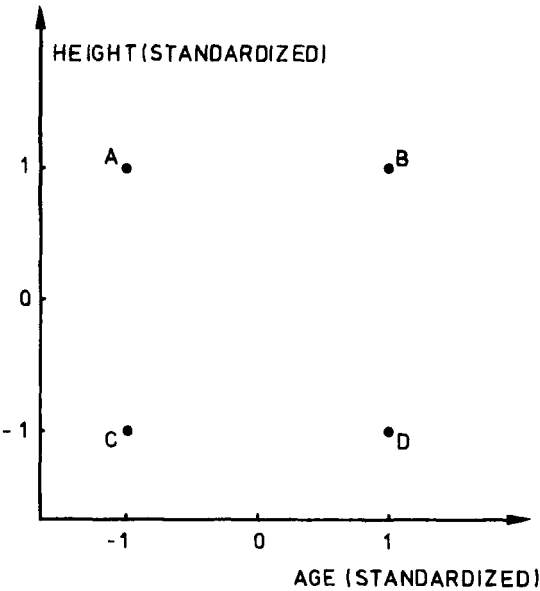


Figure 5 Standardized height versus standardized age.

consisting of a single point, or that there is only one big cluster containing four points.

From a philosophical point of view, standardization does not really solve the problem. Indeed, the choice of measurement units gives rise to relative *weights* of the variables. Expressing a variable in smaller units will lead to a larger range for that variable, which will then have a large effect on the resulting structure. On the other hand, by standardizing one attempts to give all variables an equal weight, in the hope of achieving objectivity. As such, it may be used by a practitioner who possesses no prior knowledge. However, it may well be that some variables are intrinsically more important than others in a particular application, and then the assignment of weights should be based on subject-matter knowledge (see, e.g., Abrahamowicz, 1985). On the other hand, there have been attempts to devise clustering techniques that are independent of the scale of the variables (Friedman and Rubin, 1967). The proposal of Hardy and Rasson (1982) is to search for a partition that minimizes the total volume of the convex hulls of the clusters. In principle such a method is invariant with respect to linear transformations of the data, but unfortunately no algorithm exists for its implementation (except for an approximation that is restricted to two dimensions). Therefore, the dilemma of standardization appears unavoidable at present and the programs described in this book leave the choice up to the user.

Of course, the data offered to the program may already be the result of some transformation: Often people find it useful to replace some variable by its inverse or its square, which may be more meaningful in that particular context. However, we shall assume from now on that such transformations have been performed prior to the cluster analysis.

The next step is to compute distances between the objects, in order to quantify their degree of dissimilarity. It is necessary to have a distance for each pair of objects i and j . The most popular choice is the *Euclidean distance*

$$d(i, j) = \sqrt{(x_{i1} - x_{j1})^2 + (x_{i2} - x_{j2})^2 + \cdots + (x_{ip} - x_{jp})^2} \quad (6)$$

(When the data are being standardized, one has to replace all x by z in this expression.) Formula (6) corresponds to the true geometrical distance between the points with coordinates $(x_{i1}, \dots, x_{ip})$ and $(x_{j1}, \dots, x_{jp})$. To illustrate this, let us consider the special case with $p = 2$. Figure 6 shows two points with coordinates (x_{i1}, x_{i2}) and (x_{j1}, x_{j2}) . It is clear that the actual distance between objects i and j is given by the length of the hypotenuse of the triangle, yielding expression (6) by virtue of Pythagoras' theorem. For this reason, Gower (1971b) calls (6) the *Pythagorean distance*.

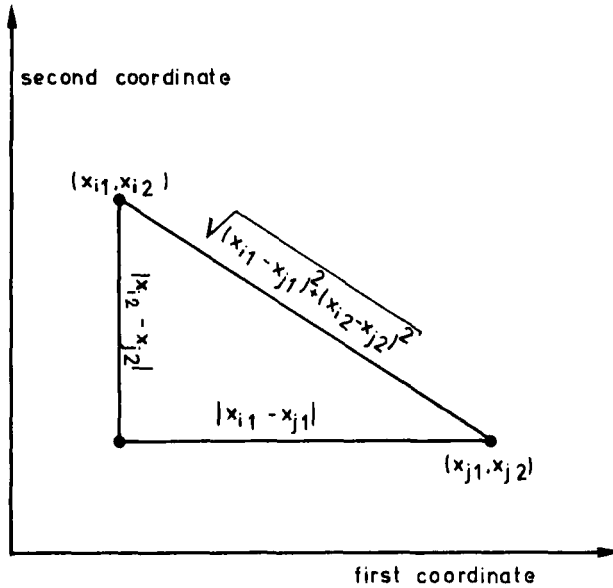


Figure 6 Illustration of the Euclidean distance formula.

Another well-known metric is the *city block* or *Manhattan distance*, defined by

$$d(i, j) = |x_{i1} - x_{j1}| + |x_{i2} - x_{j2}| + \cdots + |x_{ip} - x_{jp}| \quad (7)$$

In Figure 6, this corresponds to the sum of the lengths of the other two sides of the triangle. The Manhattan distance was used in a cluster analysis context by Carmichael and Sneath (1969) and owes its peculiar name to the following reasoning. Suppose you live in a city where the streets are all north-south or east-west, and hence perpendicular to each other. Let Figure 6 be part of a street map of such a city, where the streets are portrayed as vertical and horizontal lines. Then the actual distance you would have to travel by car to get from location i to location j would total $|x_{i1} - x_{j1}| + |x_{i2} - x_{j2}|$, corresponding to (7). This would be the shortest length among all possible paths from i to j . Only a bird could fly straight from point i to point j , thereby covering the Euclidean distance between these points. The use of the Manhattan distance is advised in those situations where for example a difference of 1 in the first variable and of 3 in the second variable is the same as a difference of 2 in the first variable and of 2 in the second.